

# Arabic-English Disparities in Large Language Model Mathematical Reasoning

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## Process Reward Evaluation, Generative Pilots, and a 200-Pair Global-MGSM Study

### Final Project Report

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Date: September 2026

Project repository: [github.com/the1ZAKK/arabic-english-llm-reasoning](https://github.com/the1ZAKK/arabic-english-llm-reasoning)

## Abstract

This study explores how mathematical reasoning systems respond to equivalent problems presented in English and Arabic. Three related studies were performed. Firstly, the Process Reward Model (PRM), Skywork-o1-Open-PRM-Qwen-2.5-1.5B, was applied to score pairs of reasoning traces that contain correct answers and controlled errors in English and Arabic languages. Secondly, two small language models – Qwen2.5-Math-1.5B-Instruct and Gemma-3-1B-IT – were tested in the 20-problem pilot in both languages. Thirdly, the 200 problems from the Global-MGSM set were used for the main experiment that was also run by two models in English and Arabic. The result of the PRM study showed that the average discrimination value between correct and wrong responses was 0.3562 in English and 0.1080 in Arabic; the paired difference was 0.2481 (95% confidence interval [0.1949, 0.3013], paired  $t = 9.76$ ,  $p < 10^{-8}$ ). In the 200-pair primary experiment, Gemma obtained 60.5% correct for English and 22.0% for Arabic, representing a 38.5 percentage-point disparity (exact McNemar  $p = 2.26 \times 10^{-17}$ ; paired bootstrap 95% CI [30.5, 46.0] pp). Qwen got 89.5% for English and 1.5% for Arabic, which represents an 88.0-point disparity ( $p = 2.09 \times 10^{-53}$ ; 95% CI [83.5, 92.5] pp). The Qwen language gap was 49.5 points higher than Gemma's (95% CI [41.5, 57.5], paired randomization  $p < .0001$ ). In addition, a bilingual semantic audit produced 14 pairs flagged as potentially problematic, with exclusion of all 14 changing the findings little. Qualitative examination also revealed that many of the Qwen Arabic errors came from solving irrelevant problems regardless of Arabic correctness, indicating that understanding and/or semantic grounding failure, rather than just computation error, is occurring in this model/experiment setup. These findings provide compelling evidence for model-specific and language-specific differences under the given experimental conditions, although it does not demonstrate that Arabic is inherently a worse reasoner than English.

**Keywords:** multilingual reasoning, Arabic, English, mathematical reasoning, process reward models, MGSM, Qwen, Gemma, language disparity

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## 1. Introduction

However, as LLMs begin to undertake more complex reasoning tasks involving mathematical computations, their perceived capability does not necessarily hold true for all languages. The machine may have the ability to do the required mathematics but may fail due to an inadequate understanding of how the problem has been formulated linguistically. This is an important point for multilingual machines as the user should not be forced to use English in order to get sound mathematical reasoning.

Word problems in mathematics provide a convenient domain for considering this problem. While the underlying numbers and operations remain constant, the surface language can be varied. GSM8K defined grade-school mathematics as the benchmark of multi-step reasoning, and MGSM has used the same paradigm in developing a benchmark for multilingual reasoning by translating some problems of GSM8K in several languages (Cobbe et al., 2021; Shi et al., 2023). In more recent multilingual benchmarks, variations in reasoning abilities between languages and benchmark translations themselves have been considered as an experimental variable (Huang et al., 2025; Peter et al., 2025).

The current work concentrates on English and Arabic languages. This experiment involves the use of three different perspectives instead of one accuracy metric. The first experiment determines whether a process reward model treats correct and incorrect reasoning the same way in English and Arabic. The second one is a smaller language-matched generative pilot that helps detect failure cases and confirms the validity of the assessment process. The third experiment is the actual experiment that involves the testing of 200 English-Arabic Global-MGSM problems against two independent generative model families (Gemma and Qwen). The third experiment is also paired in the sense that each English experiment corresponds to an Arabic experiment.

The empirical pattern is the same in all three phases. In both cases, English outperforms Arabic in terms of PRM discrimination and in direct problem-solving accuracy. However, the extent to which this holds true is very different for each of the two model categories. This point is critical since it challenges the notion of “Arabic performance” being treated as a universal constant. The difference should be seen as an interaction of multiple factors.

## 2. Research Questions and Hypotheses

### 2.1 Research questions

1. RQ1. Does a mathematical process reward model distinguish correct from incorrect reasoning equally well for matched English and Arabic inputs?
2. RQ2. Do small generative language models achieve different mathematical problem-solving accuracy on matched English and Arabic problems?
3. RQ3. Is the direction and magnitude of the English-Arabic difference stable across model families, larger samples, and benchmark-quality sensitivity analyses?
4. RQ4. When Arabic generations fail, do the failures appear primarily arithmetic, or do they also involve problem interpretation and semantic grounding?

### 2.2 Working hypotheses

- H1. Correct-versus-incorrect PRM discrimination will be larger in English than in Arabic.
- H2. Generative accuracy will be higher on English versions of matched mathematical problems than on Arabic versions.
- H3. The direction of the language-associated difference will be consistent across models, but its magnitude will be model dependent.
- H4. At least some Arabic failures will occur before calculation, through problem misunderstanding or semantic grounding failure.

**Interpretive scope.** These hypotheses concern the tested models and evaluation conditions. They are not hypotheses that Arabic itself is intrinsically less suitable for mathematical reasoning.

## 3. Related Work

### 3.1 Mathematical reasoning and verification

GSM8K was first proposed by Cobbe et al. (2021) as a dataset containing around 8,500 multi-lingual grade-school mathematics word problems for promoting reasoning ability. The same study demonstrated the effectiveness of using verifiers in selecting solutions through ranking candidates. This differentiation of generating and evaluating the generated output is critical to the current project.

Process supervision makes the evaluation procedure more detailed. The model now evaluates each step of the solution instead of giving an ultimate output label. As demonstrated by Lightman et al. (2024), process supervision can be superior to the outcome-based supervision on

the task of mathematical reasoning, and they released PRM800K as a large-scale dataset of step-level labels. The current step-level mathematics evaluation model, Skywork PRM, falls into this category, being based on Qwen2.5-Math.

### 3.2 Multilingual mathematical reasoning

The method proposed by Shi et al. (2023), called MGSM, was introduced by translating 250 GSM8K problems into several languages and showing how chain-of-thought reasoning can transfer across languages, especially at larger model scales. Later, the Global-MGSM method was introduced by unifying MGSM and some other extensions into 41 language configurations. Most importantly for this project, the English language configuration is referred to MGSM-Rev2 while the Arabic language configuration is referred to BenchMAX\_Math. Thus, the two paired rows share benchmark answers, but they are not necessarily literal translations through one translation pipeline.

The design of the BenchMAX was to evaluate multilingual abilities of large models, especially their reasoning abilities, where native speaker quality control of translations is highlighted by the creators of BenchMAX (Huang et al., 2025). Similar research highlights the potential biasing effects on multilingual benchmark gaps due to wrong translations and extractions of the correct answers. The paper by Peter et al. (2025) shows that the fixes of MGSM translations and evaluation problems can significantly affect the reported gaps between languages.

### 3.3 Why language gaps may emerge

The following are a few possible mechanisms through which the observed linguistic differences in performance could arise without necessarily indicating any inherent limitations of the language. The distribution of pretraining and tuning training corpora may not be equal among all languages, mathematical datasets may skew toward English, and language-specific tokenizers can represent similar information with varying degrees of efficiency. Petrov et al. (2023) have found major cross-linguistic differences in sequence length when using tokenizers. Another recent study of multilingual reasoning models found that failure to correctly comprehend and translate a non-English input into the dominating language in the reasoning traces can be a significant cause for performance difference (Kang et al., 2026).

### 3.4 Models examined in this project

The Qwen2.5-Math family is a specialized math model family whose technical report outlines instruction-tuned models of size 1.5B, 7B, and 72B and assesses their mathematical skills in English and Chinese languages (Yang et al., 2024). The Arab language is not mentioned as a targeted language in this report. On the other hand, Gemma 3 family is a lightweight open model family designed to have enhanced multilingual capacity and post-training to improve

mathematics, instructions, and multilingual skills (Gemma Team et al., 2025). The difference in the context of designing these two models is what makes the cross-model difference more important in this study.

## 4. Methodology

### 4.1 Experimental overview

| Experiment       | Model(s)                                  | Problems / traces                     | Paired languages | Primary outcome                            |
|------------------|-------------------------------------------|---------------------------------------|------------------|--------------------------------------------|
| PRM evaluator    | Skywork-o1-Open-PRM-Qwen-2.5-1.5B         | 20 problems $\times$ 4 response types | Arabic / English | Correct-vs-incorrect reward discrimination |
| Generative pilot | Qwen2.5-Math-1.5B-Instruct; Gemma-3-1B-IT | 20 matched problems                   | Arabic / English | Exact numerical accuracy                   |
| Main benchmark   | Same two generative models                | 200 Global-MGSM matched pairs         | Arabic / English | Paired accuracy, significance, robustness  |

Table 1. Overview of the three experimental stages.

### 4.2 Experiment 1: process reward model evaluation

Twenty math word problems were presented in matched Arabic and English versions. Four different types of responses were devised for each problem: a right answer, a wrong arithmetic solution, a wrong logical solution, and a completely wrong solution. The Skywork-o1-Open-PRM-Qwen-2.5-1.5B rewarded process rewards to the problem-solving traces. The analysis emphasized the difference between the right response and the wrong responses as a group, while preserving the three types of wrong responses in a descriptive analysis.

The main PRM performance measure was per-problem discrimination gap: the score for the right answer minus the average score of the three wrong answers. Differences between gaps for the English and Arabic versions were analyzed in each problem through paired statistics. Classification-based summary measures were also obtained but are considered auxiliary as the threshold is not portable across languages.

### 4.3 Experiment 2: 20-problem generative pilot

In order to investigate the case of direct generation with Qwen2.5-Math-1.5B-Instruct and Gemma-3-1B-IT, the same 20 pairs of math questions were provided for each of these two models. For each of these tasks, each model provided one deterministic answer for each of the languages. The numerical part of the answer was automatically extracted, and manually checked

when the automatic extraction contradicted the reasoning presented or when degeneration led to a confusing last token.

#### 4.4 Main dataset: 200 matched Global-MGSM problems

The primary dataset was sourced from CohereLabs/global-mgsm. The English and Arabic versions each had 250 test rows containing questions, answers in numbers, instructions for each language, and answer prefix. All 250 rows of each version had corresponding gold answers. According to the Global-MGSM framework, English belongs to MGSM-Rev2 and Arabic to BenchMAX\_Math. This ensures that answer alignment is highly accurate, but not necessary a direct translation.

Two-stage sampling was applied because the experiment was extended from 100 to 200 pairs after the first run. In the first stage, the first 100 source indices were sampled deterministically using Python random seed 42 from 250 aligned rows. In order to ensure that the first part remained consistent, the extension stage sampled an additional 100 source indices, using seed 42 from the other 150 rows. The result was the union of 200 source indices where the Problems 1–100 remained exactly the same while Problems 101-200 were sampled in the extension stage.

Scale of the main experiment.  $200 \text{ problems} \times 2 \text{ languages} \times 2 \text{ models} = 800 \text{ generations}$ . Each half of the experiment included 100 pairs.

#### 4.5 Models and generation settings

- Qwen/Qwen2.5-Math-1.5B-Instruct.
- google/gemma-3-1b-it.
- Deterministic generation with a maximum of 512 new tokens.
- PyTorch float16 inference with `device_map="auto"` on an NVIDIA RTX 2060 Max-Q (6 GB VRAM).
- A fixed project seed of 42; because sampling was disabled for generation, the decoding itself was deterministic.
- Language order alternated by problem to reduce systematic run-order effects.
- Only the generated completion not the prompt was decoded for answer parsing.

This evaluation deliberately employed customized symmetric prompts in English and Arabic instead of the instruction and answer\_prefix fields in Global-MGSM. This made the experiments consistent across both languages, but it implies that the results obtained here are only valid within this context. The chat format of Qwen automatically generated an English prompt for the system to reason out the answer step by step and provide it in boxed form.



## 4.6 Answer extraction and manual adjudication

The automatic extraction process normalized Arabic and Persian numerals, comma usage, decimal separators, currency indicators, and percentages. In the case of extraction, the parser favored explicit markers of final answers and Qwen boxed answers of numerical results. In cases where there were contradictory explicit final answers, the first explicit answer was chosen and the case was flagged. Last number fallback extraction was always flagged for human judgment. Comparison with gold standard involved normalizing decimal equality.

The manual adjudication process relied on a very strict rule, which was applied uniformly to all models: a model’s response was considered as correct if and only if it arrived at the gold standard benchmark as the solution to the given problem. Just citing the gold standard answer was not sufficient. However, if the model solved the problem and was then faced with repetition or loss of formatting quality, the response was considered correct. This fixed parser false negatives in both model families and, most importantly, it eliminated the Qwen Arabic false positives involving correct answer appearing while solving a different problem.

## 4.7 Qwen semantic-grounding audit

Given that Qwen generated many Arabic completions which seemed to be unrelated to the input, another audit of the prompts examined suspicious prompts at the Arabic input question, the structure of the message, rendered chat template, decoded input tokens, and completion generated. In one of the audited prompts (Problem 100 out of the first 100), the Arabic input was preserved in both rendered and tokenized form, but the generated completion contained an entirely different English word problem. This decreased the likelihood that the main source of failure was the corruption/omission of the Arabic input question in the local inference pipeline.

## 4.8 Benchmark semantic audit and sensitivity subsets

A two-language semantic analysis was done to determine the benchmark items which have either not been translated accurately into English and Arabic languages or had vague source problem. Four items were identified as definite semantic mismatches (Items 51, 100, 189, and 197); five as ambiguous translations (Items 55, 166, 171, 175, and 192); and five as ambiguous source problem (Items 43, 86, 122, 150, and 163). Minor variations in wording without affecting mathematical formulation were kept.

| Subset           | N   | Exclusions                                    |
|------------------|-----|-----------------------------------------------|
| ALL_200          | 200 | None; primary standardized benchmark analysis |
| DEFINITE_MATCHED | 196 | 4 definite semantic mismatches                |

| Subset            | N   | Exclusions                                     |
|-------------------|-----|------------------------------------------------|
| TRANSLATION_CLEAN | 191 | Semantic mismatches + 5 ambiguous translations |
| STRICT_QC         | 186 | All above + 5 ambiguous source problems        |

Table 2. Benchmark-quality sensitivity subsets.

## 4.9 Statistical analysis

Generative comparisons were matched by problem. In each of the models, the difference between English and Arabic accuracy was compared using the exact two-sided McNemar test on discordant pairs. The paired nonparametric bootstrap (10,000 resamples, seed 42) was used to compute the 95% confidence interval of the mean difference of English minus Arabic accuracy. To compare language gaps across models, each problem generated a language gap Qwen and a language gap Gemma value. Their difference was assessed with a paired bootstrap and a randomization test with 100,000 draws based on sign flips. Due to the limited resolution of the Monte Carlo test, results at the floor are reported as  $p < .0001$  instead of an exact probability.

In the PRM experiment, the difference between the discrimination gaps of English and Arabic per problem was computed using a paired t test, a Wilcoxon signed-rank test, a bootstrap confidence interval, and Cohen's dz.

## 5. Results

### 5.1 Process reward model

The PRM assigned substantially higher scores to correct English reasoning than to incorrect English reasoning, whereas the separation was smaller in Arabic. Mean scores and discrimination values are shown below.

| Language | Correct mean | Arithmetic error | Logic error | Completely wrong | Incorrect mean | Correct - incorrect gap |
|----------|--------------|------------------|-------------|------------------|----------------|-------------------------|
| Arabic   | 0.5117       | 0.3400           | 0.4466      | 0.4242           | 0.4036         | 0.1080                  |
| English  | 0.9287       | 0.5619           | 0.5698      | 0.5858           | 0.5725         | 0.3562                  |

Table 3. Skywork PRM descriptive scores on the 20 matched problems.

**Paired discrimination result.** The English-minus-Arabic difference in per-problem discrimination was 0.2481 (SD = 0.1137), with a 95% CI of [0.1949, 0.3013]. The paired t

statistic was 9.7631 ( $p = 7.7 \times 10^{-9}$ ), the Wilcoxon signed-rank  $p$  value was  $1.91 \times 10^{-6}$ , and Cohen's  $d_z$  was 2.18. All 20 problems had a larger correct-versus-incorrect gap in English.

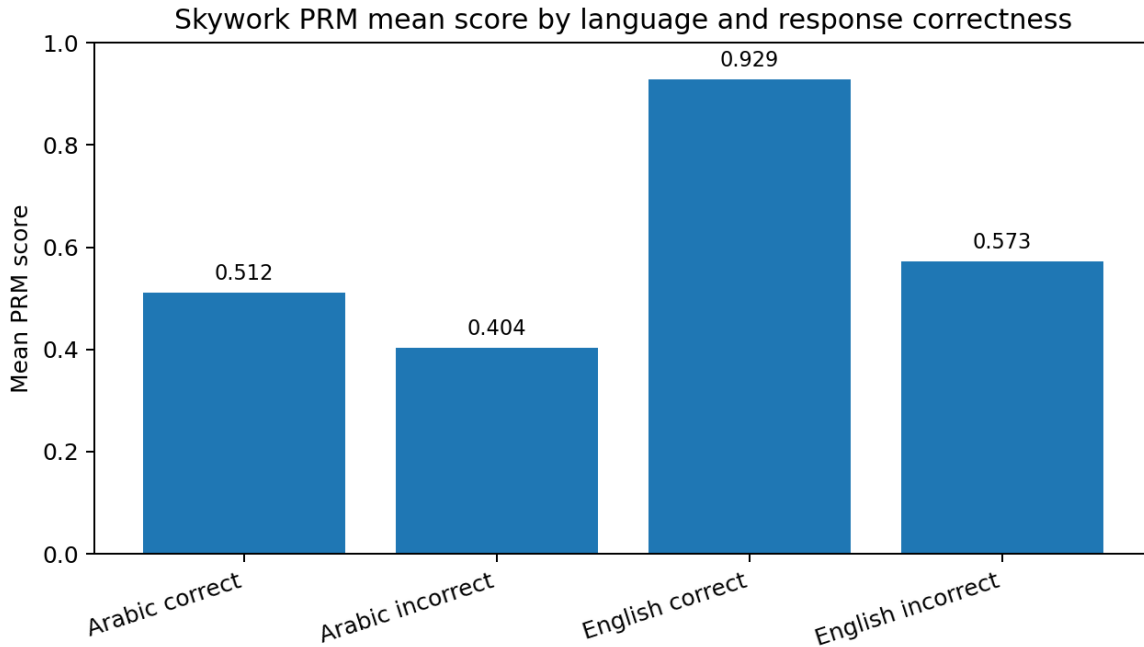


Figure 1. Mean PRM score for correct and combined incorrect reasoning by language.

At an accuracy-optimized threshold, Arabic classification accuracy was 0.7625 with AUC 0.7042, while English accuracy was 0.9875 with AUC 0.9800. These thresholded values are descriptive; the paired discrimination gap is the more direct test of whether the evaluator separates correct and incorrect reasoning equally across languages.

## 5.2 Twenty-problem generative pilot

| Model                      | Arabic      | English      | Gap    | Exact McNemar p | Paired bootstrap 95% CI |
|----------------------------|-------------|--------------|--------|-----------------|-------------------------|
| Qwen2.5-Math-1.5B-Instruct | 8/20 (40%)  | 20/20 (100%) | +60 pp | 0.000488        | 40–80 pp                |
| Gemma-3-1B-IT              | 15/20 (75%) | 20/20 (100%) | +25 pp | 0.0625          | 10–45 pp                |

Table 4. Final manually adjudicated 20-problem pilot results.

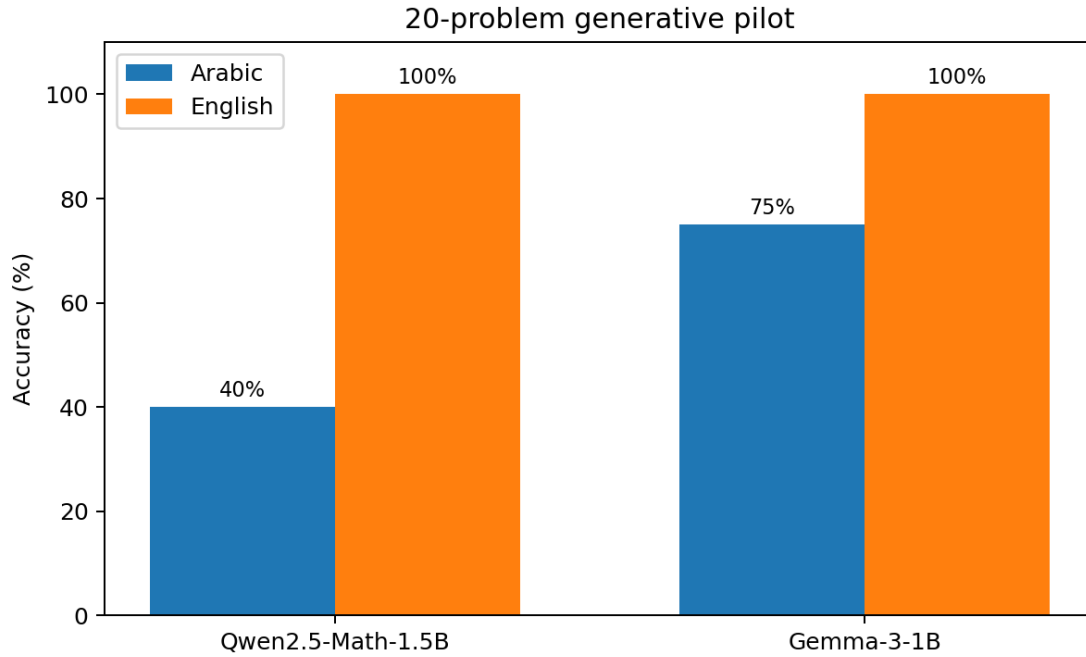


Figure 2. Accuracy in the 20-problem matched-language generative pilot.

Directional effects were created by the pilot using two independently developed models, while demonstrating that magnitude was model dependent. The Arabic mistakes of Qwen were often due to an inability to comprehend the question, but the performance of Gemma in Arabic was much better.

### 5.3 Main Global-MGSM 200-pair results

| Model                      | Arabic accuracy | English accuracy | English - Arabic | Exact McNemar p        | Paired bootstrap 95% CI |
|----------------------------|-----------------|------------------|------------------|------------------------|-------------------------|
| Gemma-3-1B-IT              | 44/200 (22.0%)  | 121/200 (60.5%)  | +38.5 pp         | $2.26 \times 10^{-17}$ | 30.5–46.0 pp            |
| Qwen2.5-Math-1.5B-Instruct | 3/200 (1.5%)    | 179/200 (89.5%)  | +88.0 pp         | $2.09 \times 10^{-53}$ | 83.5–92.5 pp            |

Table 5. Primary ALL\_200 results.

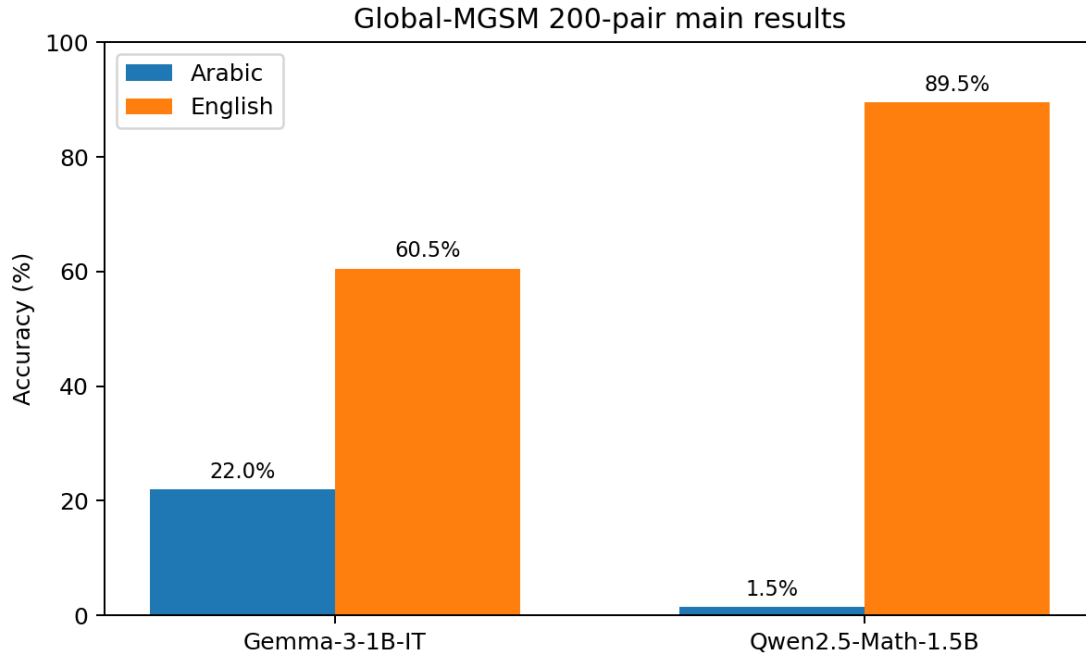


Figure 3. Accuracy on 200 matched Global-MGSM problems.

Gemma was accurate in 85 instances with English being right while Arabic was wrong, but only 8 vice versa; 36 in both languages were correct, and 71 were incorrect in both. Qwen had 176 successes only in English, no success only in Arabic, 3 instances in both languages which were correct, and 21 instances in both languages which were wrong. Therefore, the within-theory disagreements were not due to a few borderline cases.

#### 5.4 Replication across the two 100-problem halves

| Model | Problems 1–100          | Problems 101–200        | Full 200 |
|-------|-------------------------|-------------------------|----------|
| Gemma | +38 pp (23% AR, 61% EN) | +39 pp (21% AR, 60% EN) | +38.5 pp |
| Qwen  | +90 pp (2% AR, 92% EN)  | +86 pp (1% AR, 87% EN)  | +88.0 pp |

Table 6. Stability of the language gap across the staged sample expansion.

The new results did not alter the initial 100-problem pattern in any way. For Gemma, the gap was adjusted to 39 percent points from 38 percent, while for Qwen, the gap was 86 points from 90 points. Even though both the parts are derived from the same benchmark and cannot be considered independent studies, they confirm the validity of the original results.

#### 5.5 Cross-model difference in language gaps

The differences in model-specific language gaps were 49.5 percentage points, where the gap in favor of Qwen was 88.0 points while that of Gemma was 38.5 points. The 95% confidence

interval of this difference based on the paired bootstrap was [41.5, 57.5] percentage points, while the paired sign-flip randomization test produced  $p < .0001$ . In Arabic, both models were right on only 2 questions; Gemma was right alone on 42, Qwen alone on 1, and neither on 155. In English, both were right on 117, Gemma alone on 4, and Qwen alone on 62.

## 5.6 Benchmark-quality robustness

| Subset            | N   | Gemma gap | Qwen gap  | Qwen - Gemma gap |
|-------------------|-----|-----------|-----------|------------------|
| ALL_200           | 200 | +38.50 pp | +88.00 pp | +49.50 pp        |
| DEFINITE_MATCHED  | 196 | +37.76 pp | +88.27 pp | +50.51 pp        |
| TRANSLATION_CLEAN | 191 | +38.22 pp | +88.48 pp | +50.26 pp        |
| STRICT_QC         | 186 | +38.71 pp | +88.17 pp | +49.46 pp        |

Table 7. Robustness of the language-associated accuracy gaps to benchmark-quality exclusions.

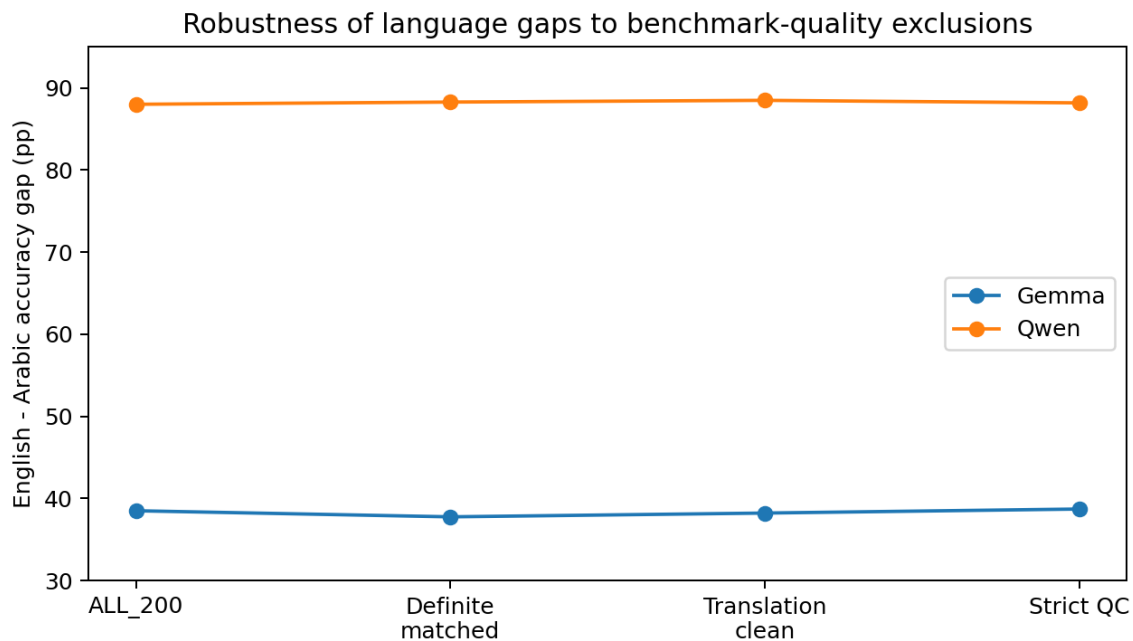


Figure 4. Language gaps remain stable under progressively stricter benchmark-quality filtering.

The strictest sensitivity analysis removed all 14 identified semantic mismatches, ambiguous translations, and ambiguous source problems, leaving 186 matched pairs. Gemma then achieved 23.1% Arabic and 61.8% English accuracy, a 38.71-point gap (McNemar  $p = 4.60 \times 10^{-16}$ ; bootstrap 95% CI [30.65, 46.77] pp). Qwen achieved 1.61% Arabic and 89.78% English, an 88.17-point gap ( $p = 8.55 \times 10^{-50}$ ; CI [83.33, 92.47] pp). The strict-QC cross-model gap

difference was 49.46 points (95% CI [40.86, 58.06], randomization  $p < .0001$ ). The conclusions therefore do not depend on the 14 identified questionable items.

### 5.7 Qualitative Qwen Arabic failure pattern

The manual analysis identified a particular type of Qwen failure mode. It turned out that many of the Arabic outputs did not make just a calculation error; rather, they came up with an English explanation of a totally different task. For instance, an Arabic problem related to cost, journey, or counting could result in an output related to books, investment, probability, or another completely new task using part of the input numbers. This is why a mere numerical extraction was dangerous because, in several cases, the new task happened to give the correct gold number of the benchmark.

The semantic grounding check helped to confirm this hypothesis. In suspicious examples, the original Arabic question stayed the same in the message structure, the Qwen chat template, and the decoded input token sequence. Yet, the model still produced a new task. Of course, this does not prove any concrete internal process but gives evidence that the failures observed were indeed the failures of understanding/grounding.

## 6. Discussion

### 6.1 Answer to the research questions

RQ1 is supported for the tested PRM: The reward model discriminates between good and bad reasoning far more effectively in English than in Arabic. RQ2 is supported for both generative model families, and for both the pilot study and main benchmarking study. RQ3 is likewise supported: The direction is reliably in favor of English, though the magnitude is extremely inconsistent, ranging from about 39 percentage points for Gemma to 88 percentage points for Qwen in the main study. RQ4 is supported by description for Qwen: Many of the Arabic failures were due to an inability to understand or ground the problem prior to mathematical reasoning.

### 6.2 Why the cross-model magnitude matters

If the Arabic language in itself imposes one “reasoning penalty,” then we should expect similar results from various families of models. However, the difference between Qwen and Gemma for the same set of 200 paired questions was significant. Qwen performed significantly better than Gemma in English (89.5% versus 60.5%) but significantly worse in Arabic (1.5% versus 22.0%). Such an interaction proves that performance on languages cannot be separated from the training and specialization of models. Thus, it can be stated that the result is model-to-language performance difference rather than the general Arabic deficiency.

The information about the models gives important additional context. While Qwen2.5-Math is advertised as a math model in English and Chinese, Gemma 3 is a multilingual model. However, this does not explain the experimental results in itself since there are many other factors which differ between these two models.

### 6.3 Evaluation models can also be language sensitive

As is demonstrated by the PRM experiment, multilingual bias is not just an issue of generation. If a reward model used to rank or verify reasoning abilities happens to be less discriminating in one of the languages, this could have practical implications for best-of-N, rejection sampling, reinforcement learning, and quality control tasks. That is, in a multilingual generation pipeline, the bias introduced by the process reward model could actually get exacerbated because such models were meant to evaluate intermediate reasoning abilities, not just the end number.

### 6.4 Translation and evaluation artifacts

Some recent research highlights the danger of overstating multilingual math achievement disparities due to translation errors and scoring of answers (Peter et al., 2025). Both issues arose explicitly in the current investigation. The bilingual review revealed unambiguously that there were semantic discrepancies and ambiguous questions, while the automatic parsing sometimes scored degenerated answers incorrectly. However, the ultimate findings did not change despite a rigorous manual evaluation and elimination of all 14 benchmark-quality issues identified. It does not mean that there are no other translation-related problems left, yet it significantly diminishes the probability that the huge effect sizes are caused only by them.

### 6.5 Interpreting Qwen’s semantic-grounding failures

The Arabic behavior exhibited by Qwen is reflective of an error occurring prior to task interpretation. In this case, numerals and fragments of structure are often recycled, while the semantic scenario is updated. As per recent studies on multilingual reasoning, it is believed that an error in understanding can be the cause of very large differences across languages in reasoning performance (Kang et al., 2026). This is because it becomes difficult to map non-English inputs into the representation required for reasoning. While this is one plausible explanation of the observations made, it does not constitute a mechanistic test.

### 6.6 Practical implications

- Multilingual evaluation should test reasoning per language independently and not rely on benchmark scores from English.
- Incentive mechanisms and judges must be tested multilingually because a good English verifier is not necessarily a good Arabic verifier.



- Pair benchmark testing is great, but we cannot simply assume good translation.
- The extraction of answers is part of the evaluation process and should be checked when outputs fail or have contradicting answers.
- Models chosen for reasoning in Arabic should depend on multilingual training and not just English math benchmark scores.

## 7. Limitations and Threats to Validity

**Model scope:** Only two small generative model families and one 1.5B PRM were successfully evaluated. The findings should not be generalized to all LLMs, frontier models, or all versions of Qwen and Gemma.

**Qwen specialization:** The Qwen2.5-Math technical report explicitly centers English and Chinese. Its poor Arabic performance is therefore evidence about this tested model under Arabic input, not evidence that the wider Qwen family is incapable of Arabic reasoning.

**Benchmark exposure and contamination:** MGSM is derived from GSM8K, a widely used benchmark. The study cannot determine whether benchmark items, close paraphrases, or related English mathematical data appeared during pretraining or post-training. Differential exposure could inflate English accuracy or widen language gaps.

**Cross-source bilingual pairs:** Global-MGSM uses different source pipelines for English and Arabic. Even though all corresponding gold answers matched and a semantic audit was performed, the pairs are not guaranteed to be perfect translations.

**Prompting choice:** Custom symmetric prompts were used instead of the dataset-provided instruction and answer prefix. Qwen also inserted an English system instruction through its chat template. Results are therefore conditional on this prompting pipeline.

**Single deterministic generation:** Each model produced one deterministic completion per language/problem. The study measures this decoding condition rather than the distribution of outcomes under stochastic sampling.

**Manual adjudication:** Human judgment was required for parser failures, conflicting final answers, response degeneration, and semantic false positives. A strict rule was applied consistently, but adjudication still introduces potential subjectivity.

**Chain-of-thought interpretation:** Visible reasoning text is behavioral output, not direct access to the model's internal computation. Descriptions such as “problem-understanding failure” refer to the observed response pattern.

**Hardware constraint:** A larger Qwen2.5-Math-PRM-7B evaluator was attempted but could not be run reliably within the available 6 GB GPU / host-memory constraints. The PRM conclusions therefore rely on the Skywork 1.5B evaluator.

**Arabic diversity:** The project evaluates Modern Standard Arabic benchmark text, not dialectal Arabic, code-switching, or all Arabic scripts and domains.

## 8. Conclusion

In both process/reward evaluation and math problem solving, the evaluated systems exhibited significant language differences between the matching English and Arabic problems. The difference was consistent – English was always better, but it varied significantly depending on the model. In the 200 pairs Global-MGSM evaluation, Gemma-3-1B-IT achieved 60.5% accuracy for English and 22.0% for Arabic, and Qwen2.5-Math-1.5B-Instruct 89.5% for English and 1.5% for Arabic. The gap between the two models in terms of language was close to 50 percentage points.

These results stayed almost the same after exclusion of every benchmark sample that was identified as a semantic mismatch or ambiguous translation or source math problem. The 20 problems pilot confirmed the same trend, and the PRM experiment has shown that the evaluation quality can depend on the language itself. Examination of Qwen showed that a large part of the problems with Arabic were related to grounding or problem understanding and not arithmetic calculation.

The appropriate conclusion, however, is limited yet significant: with the models used and benchmarked on, prompts and decoding settings provided, mathematical reasoning performance was significantly better in English compared to Arabic, and the magnitude of the difference between the two languages relied heavily on the model. The research does not provide any basis for stating that Arabic language is inherently detrimental to the process of reasoning. Rather, it proves the importance of multilingual evaluation of the models.

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## Appendix A. Reproducibility Summary

### A.1 Software and hardware

- Windows environment with Python 3.11.9.
- PyTorch 2.11.0+cu128 with CUDA available.
- NVIDIA RTX 2060 Max-Q with 6 GB VRAM; approximately 15.8 GB system RAM.

- Transformers 5.16.1; Hugging Face Hub 1.29.0; Accelerate and bitsandbytes installed.
- Hugging Face caches relocated to drive D: during model downloads.

## A.2 Key generated artifacts

- global\_mgsm\_ar\_en\_200.jsonl — final 200-pair matched dataset.
- gemma\_mgsm\_200\_adjudicated.csv and qwen\_mgsm\_200\_adjudicated.csv — final response-level results.
- gemma\_mgsm\_200\_paired\_results.csv and qwen\_mgsm\_200\_paired\_results.csv — paired problem-level results.
- mgsm\_200\_model\_summary.csv — final within-model statistics.
- mgsm\_200\_cross\_model\_summary.csv — cross-model language-gap comparison.
- mgsm\_200\_qc\_exclusions.csv and mgsm\_200\_robustness\_summary.csv — bilingual QC and sensitivity analysis.
- qwen\_arabic\_prompt\_audit.txt — prompt/token audit of suspicious Arabic Qwen failures.
- Pilot and PRM CSVs/scripts retained in the same repository for full experimental history.

## A.3 Repository

The full experiment, scripts, raw outputs, adjudicated results, and robustness analyses are available at: <https://github.com/the1ZAKK/arabic-english-llm-reasoning>

The report was prepared after the completed MGSM-200 experiment was committed as: **3e4070e** — **Complete 200-pair Arabic-English MGSM experiments and robustness analysis**

## A.4 Report-ready headline result

**Primary 200-pair finding:** Gemma-3-1B-IT scored 60.5% in English versus 22.0% in Arabic (gap +38.5 pp), while Qwen2.5-Math-1.5B-Instruct scored 89.5% in English versus 1.5% in Arabic (gap +88.0 pp). Both within-model gaps were highly significant, the Qwen gap exceeded the Gemma gap by 49.5 pp (95% CI 41.5–57.5 pp; paired randomization  $p < .0001$ ), and the result was essentially unchanged under strict benchmark-quality filtering.